Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, ODD Semester, 2026-2027

Assignment-9 : D'Alembert's solution, Wave and Heat Equations

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1. Derive D'Alembert's solution for the wave equation

$$u_{tt} - c^2 u_{xx} = 0, \qquad x \in \mathbb{R},\ t > 0.$$

Hence compute the solution of Cauchy Problems:

(a) $u(x,0) = \sin x$, $u_t(x,0) = \cos x$, (b) $u(x,0) = e^{-|x|}$, $u_t(x,0) = x^2$,

(c) $u(x,0) = x$, $u_t(x,0) = 4cx$, (d) $u(x,0) = x(1-x)$, $u_t(x,0) = \sin \pi x$.

2. Find the D'Alembert's solution of the following initial boundary-value problem:

(a) $u_{tt} - 4u_{xx} = 0,\ x > 0,\ t > 0,$
$u(x,0) = |\sin x|$, $u_t(x,0) = 0$, $x > 0$; $u(0,t) = 0$, $t \geq 0$.

(b) $u_{tt} - u_{xx} = 0,\ x > 0,\ t > 0,$
$u(x,0) = \cos \dfrac{\pi x}{2}$, $u_t(x,0) = 0$, $x \geq 0$; $u_x(0,t) = 0$, $t \geq 0$.

(c) $u_{tt} - 9u_{xx} = 0,\ x > 0,\ t > 0,$
$u(x,0) = 0$, $u_t(x,0) = x^3$, $x \geq 0$; $u(0,t) = 0$, $t \geq 0$.

(d) $u_{tt} - c^2 u_{xx} = 0,\ x > 0,\ t > 0,$
$u(x,0) = e^{-x}$, $u_t(x,0) = e^x$, $x \geq 0$; $u(0,t) = 0$, $t \geq 0$.

3. Use the method of separation of variables to solve the following problems:

(a) $y^3 u_x + x^2 u_y = 0,\ u(x,0) = \exp(x^3/3)$,

(b) $u_x = 2u_t + u,\ u(x,0) = 6\exp(-3x)$,

(c) $u_x + 3u_y = 0,\ u(0,y) = 4\exp(-2y) + 3\exp(-6y)$,

(d) $4u_x + u_y = 3u,\ u(0,y) = 3\exp(-y) - \exp(-5y)$,

(e) $u_{xx} = u_y + 2u,\ u_x(0,y) = 1 + \exp(-3y),\ u(0,y) = 0$,

Ans: (a) $u(x,y) = e^{\frac{x^3}{3} - \frac{y}{4}}$. (b) $u(x,t) = 6e^{-3x-2t}$, (c) $u(x,y) = 4e^{6x-2y} + 3e^{18x-3y}$,
(d) $u(x,y) = 3e^{x-y} - e^{2x-5y}$, (e) $u(x,y) = \dfrac{1}{2\sqrt{2}}\left[e^{\sqrt{2}x} - e^{-\sqrt{2}x}\right] + e^{-3y}\sin x$.

4. Solve the following by method of separation of variables:

(a) $u_{tt} - c^2 u_{xx} = 0,\ 0 < x < L,\ t > 0,$
$u(x,0) = \cos \dfrac{\pi x}{L}$, $u_t(x,0) = 0$, $0 \leq x \leq L$,
$u(0,t) = 0$, $u(L,t) = 0$, $t \geq 0$.

(b) $u_{tt} - c^2 u_{xx} = 0,\ 0 < x < L,\ t > 0,$
$u(x,0) = 0,\ u_t(x,0) = 5\cos\dfrac{2\pi x}{L}\sin\dfrac{3\pi x}{L},\ 0 \le x \le L,$
$u(0,t) = 0,\ u(L,t) = 0,\ t \ge 0.$

(c) $u_{tt} - 4u_{xx} = 0,\ 0 < x < 1,\ t > 0,$
$u(x,0) = 0,\ u_t(x,0) = x(1-x),\ 0 \le x \le 1,$
$u(0,t) = 0,\ u(1,t) = 0,\ t \ge 0.$

(d) $u_t - 4u_{xx} = 0,\ 0 < x < 1,\ t > 0,$
$u(x,0) = x^2(1-x),\ 0 \le x \le 1,$
$u(0,t) = 0,\ u(1,t) = 0,\ t \ge 0.$

(e) $u_t - u_{xx} = 0,\ 0 < x < 2,\ t > 0,$
$u(x,0) = x,\ 0 \le x \le 1,$
$u(0,t) = 0,\ u_x(2,t) = 1,\ t \ge 0.$

(f) $u_t - u_{xx} = 0,\ 0 < x < 100,\ t > 0,$
$u(x,0) = 60,\ 0 < x < 50$ and $u(x,0) = 40,\ 50 < x < 100,$
$u_x(0,t) = 0,\ u_x(100,t) = 1,\ t \ge 0.$

(g) $u_t - ku_{xx} = 0,\ 0 < x < 1,\ t > 0,$
$u(x,0) = x(1-x),\ 0 \le x \le 1,$
$u(0,t) = 2,\ u(1,t) = 3,\ t \ge 0.$

5. A tightly stretched string of length l has its ends fastened at $x = 0$ and $x = l$. The mid point of the string is then taken to height h and then released with initial velocity zero. Find the displacement of the string at every x and at every time t.

   Ans:
   $$u(x,t) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{l}\right)\cos\left(\frac{n\pi ct}{l}\right), \qquad c_n = \frac{8h}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right).$$

6. If a string of length l is initially at rest in the equilibrium position and each of its points is given the velocity $\sin^3(\pi x/l)$. Find the displacement of the string at every x and every t.

   Ans:
   $$u(x,t) = \left(\frac{3l}{4\pi c}\sin\frac{\pi x}{l}\right)\sin\left(\frac{\pi ct}{l}\right) - \left(\frac{l}{12\pi c}\sin\frac{3\pi x}{l}\right)\sin\left(\frac{3\pi ct}{l}\right).$$

7. A uniform string of length l, constant density ρ and tension T_0 has its equilibrium position along the x-axis, with end at $x = 0$ fixed and the end at $x = l$ free to move (in a plane) perpendicular to the x-axis in such a way that the end of the string remains parallel to the x-axis. If $u(x,t)$ is the transverse displacement of the string at a point x and time $t > 0$, and released from rest while in the displaced position $u(x,0) = hx(l-x)^2$, where $h > 0$ is small, find the subsequent motion of the string.

   Ans:
   $$u(x,t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{(2n-1)\pi x}{2l}\right)\cos\left(\frac{(2n-1)\pi ct}{2l}\right), \qquad c^2 = T_0/\rho,$$

$$a_n = \frac{32hl^3}{\pi^4}\left[\frac{6(-1)^n + (4n-2)\pi}{(2n-1)^4}\right], \qquad n = 1, 2, 3, \ldots$$

8. A tightly stretched string with fixed end points $x = 0$ and $x = 1$ initially in a position given by

$$f(x) = \begin{cases} x, & 0 \le x \le \dfrac{1}{2} \\ 1 - x, & \dfrac{1}{2} \le x \le 1. \end{cases}$$

If it is released from this position with velocity a, a constant, perpendicular to x-axis, find the displacement at every x and every time t.

Ans: $u(x,t) = \displaystyle\sum_{n=1}^{\infty} \sin(n\pi x)\big[a_n \cos(n\pi ct) + b_n \sin(n\pi ct)\big]$,

where $a_n = \left(\dfrac{4}{n^2\pi^2}\right)\sin\left(\dfrac{n\pi}{2}\right)$, $\quad b_n = \left(\dfrac{2a}{n^2\pi^2 c}\right)\big[1 + (-1)^{n+1}\big]$, $\quad n = 1, 2, 3, \ldots$

9. The points of trisection of a string are pulled aside through the same distance on opposite sides of the position of equilibrium, and the string is released from rest. Compute the subsequent wave motion of the string, and show that the mid-point of the string always remains constant.

10. A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and the temperature initially is

$$f(x) = \begin{cases} x, & 0 \le x \le 50 \\ 100 - x, & 50 \le x \le 100. \end{cases}$$

Find the temperature $u(x,t)$ at every x and every time t.

Ans:

$$u(x,t) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{l}\right)\exp\left(-\frac{n^2\pi^2c^2t}{l^2}\right), \qquad c_n = \frac{4l}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)$$

11. A metal bar of length l has its ends $x = 0$ and $x = l$ insulated. Initially the bar is at temperature given by $6 + 4\cos\left(\dfrac{3\pi x}{l}\right)$, $0 < x < l$. Assuming the surface of the bar is insulated, find the temperature everywhere in the bar at time t.

Ans:

$$u(x,t) = 6 + 4\cos\left(\frac{3\pi x}{l}\right)\exp\left(-\frac{n^2\pi^2c^2t}{l^2}\right)$$

12. Solve

$$u_t = u_{xx}, \quad 0 < x < 10,\ t > 0$$

$$u(0,t) = 90,\ t \ge 0$$

$$u(10,t) = 60,\ t \ge 0$$

$$u(x,0) = 5x + 50, \quad 0 < x < 10.$$

Ans:

$$u(x,t) = -3x + 90 + \sum_{n=1}^{\infty} c_n \exp\left(-\frac{n^2\pi^2c^2t}{100}\right)\sin\left(\frac{n\pi x}{10}\right), \qquad c_n = \frac{80}{n\pi}\big[(-1)^{n+1} - 1\big],\ n = 1, 2, 3, \ldots$$

13. The ends A and B of a rod [20 cm long] are maintained at 30^0C and 80^0C, respectively until steady state prevails. The temperatures of the ends are then suddenly changed to 40^0C and 60^0C, respectively. Find the temperature distribution in the rod at time t.

    Ans

$$u(x,t) = x+40+\sum_{n=1}^{\infty} c_n \sin\Big(\frac{n\pi x}{l}\Big) \exp\Big(-\frac{n^2\pi^2c^2t}{l^2}\Big), \qquad c_n = \frac{20}{n\pi}\big[2(-1)^{n+1}-1\big],\ n = 1,2,3,\ldots$$

14. The temperature at one end of a bar of 50 cm long is kept at 0^0C and other end is kept at 100^0C until steady state conditions prevail. The two ends of the bar are then suddenly insulated. Find the temperature distribution in the rod at time t.

    Ans: $l = 50$ and

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos\Big(\frac{n\pi x}{l}\Big) \exp\Big(-\frac{n^2\pi^2c^2t}{l^2}\Big), \qquad A_0 = l,\ A_n = \frac{4l}{n^2\pi^2}(\cos n\pi-1),\ n = 1,2,3,\ldots$$

15. Find the temperature distribution in a rod of length π, one end of which is kept at zero temperature and the other end of which loses heat at a rate proportional to the temperature at that end $x = \pi$. The initial temperature distribution is given by $f(x) = x$.

16. A bar AB of length 10 cm has its ends A and B kept at $20°$C and $40°$C temperature respectively until a steady state is reached. Then, the temperature at A is increased to $50°$C, and that at B reduced to $10°$C, and these temperatures are maintained. Find the subsequent temperature distribution in the bar. Show that the temperature at the middle point of the bar remains unaltered for all time.

17. A bar of length 30 cm has its ends kept at $20°$C and $80°$C temperature respectively until a steady state is reached. Then, the temperature at each end suddenly reduced to $0°$C, and these temperatures are maintained thereafter. Find the subsequent temperature distribution in the bar.

18. A rod of 100 cm length has its end kept at $0°$C and $100°$C temperature respectively until a steady state condition prevails. The two ends are suddenly insulated and maintained so. Find the subsequent temperature distribution in the rod. Show that the sum of temperatures at any two point equidistant from the center is always $100°$C.